PREDICTIVE PERSONNEL WELFARE MONITORING · SIH 2026 · PS 26186

How SAMVEDNA decides

Every number on this page came out of the system running on your machine just now. Nothing here is illustrative.

config 2026.09.05-1   597 tests   8 languages   7.7 ms per person

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THE PROBLEM

A risk score is not a decision

Published machine-learning models for burnout risk reach roughly **AUROC 0.72** on personnel-record data. Deploy a raw score like that across ten lakh personnel and you name thousands of people wrongly — each one a soldier told, in effect, that a computer thinks something is wrong with them.

The problem statement names its own hardest risks: false positives, stigmatisation, and trust. A bare classifier makes all three worse. So the score is not allowed to be the answer.

> The model raises the concern. The evidence must corroborate it. *The arithmetic makes the decision.* A welfare officer takes the action.

THE MECHANISM

Four gates, and all four must clear

A model can suggest that somebody is struggling. Before that suggestion becomes a name on an officer's screen, it has to pass four separate arithmetic tests. Each one asks a different question, and each is computed by hand-checkable formula — not by a model.

`evidence`	Is there enough corroborated signal to be talking about a person at all? One domain alone can never clear this — the maths won't allow it.	≥ 0.65
`consistency`	Do the signals agree with each other, or is one anomaly shouting? Benign explanations — a unit deployment, sanctioned leave — push this down.	≥ 0.70
`persistence`	Sustained, or a bad week? Weighted toward the 30-day window: long enough to exclude a rough patch, short enough to act before a crisis.	≥ 0.60
`actionability`	Is there a welfare action that fits, that isn't already being taken, that this person consented to? Three hard vetoes — any one forces zero.	≥ 0.50

WATCH IT WORK — CASE 1 OF 2

All four clear, so a name is produced

A real case from last night's run. Three domains deviating, sustained over months, a voluntary self-assessment among them, and nothing in the unit explaining it away.

1a120eaf3d… composite 0.905 **ESCALATE**

evidence 0.754 CLEAR

```
evidence = 0.50 × 0.708 + 0.30 × 0.667 + 0.20 × 1 = 0.754
```

Volume, breadth, and a validated self-assessment present.

consistency 1.000 CLEAR

```
consistency = mean(1.000, 1.000) = 1.000
```

The domains agree, and no benign explanation was found.

persistence 0.943 CLEAR

`0.20 × 1.000 + 0.45 × 1.000 + 0.35 × 0.837 = 0.943`

Breaching almost every day across 7, 30 and 90-day windows.

actionability 1.000 CLEAR

`actionability = 1 × 1 × 1 × (0.75 + 0.25 × 1) = 1.000`

Action matched · not a duplicate · consented · capacity available.

A name is released to the welfare officer.

Recommended: WLF-COUNSEL — referral to the unit counselling cell.

The disclosure is written to the audit ledger *before* the name is looked up.

WATCH IT REFUSE — CASE 2 OF 2

One gate misses by 0.006, and nobody is named

Same machinery, different person. This jawan shows a strong, sustained deviation across three domains — evidence **1.000**, persistence **0.965**. On any dashboard built around a risk score, this is a flag.

But 45–61% of their unit is deviating the same way, because the unit is deployed. The Confounder Check finds it, contributes mass against the signal, and consistency lands at **0.694** against a **0.700** threshold.

pid-mon-0001 composite 0.817 MONITOR · NO NAME

evidence 1.000 CLEAR

`evidence = 0.50 × 1.000 + 0.30 × 1.000 + 0.20 × 1 = 1.000`

consistency 0.694 HOLD

`consistency = mean(0.667, 0.708, 0.708) = 0.694`

Unit-wide operational tempo on all three domains — mass 0.70 each.

persistence 0.965 CLEAR

```
0.20 × 1.000 + 0.45 × 1.000 + 0.35 × 0.900 = 0.965
```

actionability 1.000 CLEAR

```
actionability = 1 × 1 × 1 × (0.75 + 0.25 × 1) = 1.000
```

No name is produced. No alert is raised.

The system instead says what would change its mind, and the sentence is computed by inverting the gate — never written by a model:

"the duty_roster deviation is still present after the unit's current deployment cycle ends (op-tempo confounder, 0.70)"

This is the moment the whole system exists for.

A unit-wide pattern is a **command workload problem**, not an individual welfare concern. Naming this person would have been both wrong and the fastest way to lose the force's confidence. The system stays silent — and it stays silent because the arithmetic said so, not because anybody wrote a rule for the demo.

ONE NIGHT, END TO END

What actually runs at 02:00

Seven stages. Each has its own timing, row counts and status, so a failure is visible rather than silent. These are last night's real figures.

01	**Ingest**	Seven service-record connectors pull leave, roster, deployment, workload, transfer, training, self-report.	456,841 rows
02	**Protect**	Service number → pseudonym, then consent filter. **No identifier goes past this line.**	456,841 rows

03	Features	Personal and unit baselines over 7 / 30 / 90 / 180 days.	240 people
04	Deviate	Who has moved from *their own* baseline — not from a fixed threshold everyone is measured against.	128 candidates
05	Score	The risk model runs. If it's missing, the night continues — the gates don't need it.	128 scored
06	Review	Three reviewers argue in parallel: the case for, the case against, and what was tried before.	128 reviewed
07	Gate	Four gates, then the verdict. **This is the only stage that decides anything.**	3 escalated

125 people were watched and not named. That is not the system failing to find them — it is the system declining to accuse them on evidence that didn't hold up.

DOES IT SCALE

7.7 ms per person, and one honest problem

Measured over cohorts from 240 to 4,000 people — 7.6 million records at the top end — and reproducible with `tools/benchmark.py`. Per-person cost *falls* as the cohort grows, so the extrapolation is arithmetic rather than optimism: **ten lakh personnel is 2.1 hours on one core, 16 minutes on eight.** One commodity server, no GPU, no distributed system, because every person is scored independently.

And the half that is not good news

The escalation rate is stable near **0.8%**. At 240 people that is 3 cases and a good night's work. At ten lakh it is about **8,200 cases a night** — roughly 1,370 welfare officers at six conversations each. That is not a deployable number.

It is also not a compute problem. It is an operating-point decision, and the system exposes the dial rather than hiding it: the evidence threshold at 0.65 gives ~1.20%, at 0.70 gives ~0.65%, at 0.80 gives ~0.05%. A deployment sets it against the size of its welfare cell. Quoting the compute figure without this one would be exactly the overclaim the gates exist to prevent.

THE DISCLOSURE BOUNDARY

Who can see what, and who never can

This is enforced in one layer of the software, not in the interface. A bug in the console cannot leak a name, because the API never returned one.

The jawan

+ Their own data, their own drivers, their own consent — in their own language.

– Anyone else's anything.

Welfare officer

+ Identity, drivers and a recommended action — only for escalated cases in their own unit.

– Raw questionnaire answers. Anyone under MONITOR. Other units.

Commanding officer

+ Unit aggregates only, covering at least five people, with noise added.

– Any individual, ever. There is no drill-down and no API that would answer one.

Auditor

+ The full hash-chained ledger — every access, verdict, override and consent change.

– The content of anything. Only that an event occurred.

Two rules that sound small and aren't

A failed ledger write aborts the disclosure. The audit record is written before the name is looked up, so there is no ordering in which a name exists without a record of it.

Withdrawing consent is never reported. Not to the commanding officer, not to the welfare officer. There is exactly one way to read revocation events and it is named `revocations_for_auditor` — no count, no rate, no per-unit view exists for a dashboard to group by.

CLEARING SOMETHING UP

What TrueVoice has to do with this: nothing

TrueVoice is a different project — real-time voice biomarkers during GP consultations, built at a London hackathon. **None of it is in SAMVEDNA.** Zero imports, zero shared files, zero shared code.

PROJECT	WHAT IT IS	USED IN SAMVEDNA?
TrueVoice	Real-time voice biomarkers and clinical transcript concordance during GP consultations.	No — nothing at all
Prosodia	Clinical voice concordance with a pure decision core and exact event replay.	No code — architectural patterns only, rewritten from scratch
SAMVEDNA	This system. 64 Python modules, 22 web modules, 597 tests, 8 languages.	Stands alone

What carried over from Prosodia was *how to build*, not what was built: keep the deciding layer pure and free of I/O so an auditor can interrogate it; make evidence carry its own numbers so a reader can check the reasoning; report degraded stages instead of faking them. Those are habits a person moves between projects. Every line here is new.

This is enforced, not asserted: the build fails on any import from a neighbouring project, and the archive is unpacked into an unrelated directory and run cold as part of the test suite.

TRY IT YOURSELF

Three things worth doing in the running system

1 · Open a case and click a gate bar at `/officer`. The formula opens with the real numbers in it. That is what turns “the computer says 0.694” into something an officer can defend to the person sitting opposite them.

2 · Switch to UNIT-02 at `/unit` while scoped to UNIT-01. You get a refusal, because scope is enforced in the disclosure layer rather than by hiding a button.

3 · Pick हिन्दी at `/me`, set your toggles, and try to save. It refuses — that translation has not been read by a native speaker, and the system will not record consent against words nobody has verified.

Every figure on this page was read from the running system, not written by hand.

Console `localhost:3100` · API `localhost:8090` · build log `docs/PHASES.md`

Welfare support, never disciplinary action. Outputs are barred in software from ACR, promotion, posting and disciplinary processes.